

Contact

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THIS NOTEBOOK IS YOURS. USE IT TO SHARE YOUR NOTES

Part 0: About operating systems (and Linux)

Prerequisite resources

- Introduction to Operating Systems: [Operating Systems I. Introduction Ludovic Apvrille ludovic.apvrille@telecom-paris.fr](#) Eurecom, office 470
- UNIX 101: https://www.opsschool.org/unix_101.html
- The Shell: <https://missing.csail.mit.edu/2020/course-shell/>

Part 1: *Versioning all your contributions with Git*

1. Monday 7th September 2026 @ 8:00 AM

Course material

1. Wikipedia on Git: <https://en.wikipedia.org/wiki/Git>
2. Video about Git: <https://youtu.be/hwP7WQkmECE>
3. About [SHA1](#) and the anatomy of a commit: <https://blog.thoughtram.io/git/2014/11/18/the-anatomy-of-a-git-commit.html>
4. Git cheat sheet: <https://education.github.com/git-cheat-sheet-education.pdf>
5. "Oh shit, Git!"
6. Illustrated tips by Julia Evans: <https://jvns.ca/>

Homework

If you want to use your own computer/laptop, please install the following tools before the next session:

- [Git](#)

- [LaTeX](#)
- [Python](#)
- [Marimo](#)

Evaluation

- 1 Small project with LaTeX (A)
- 1 Project with GitHub Pages (B)
- 1 Project with Python (C)
- 1 Big Project with Marimo (D)

$$\text{Final Grade(FG)} = (A+B+C+2xD5)/5, \text{ if } FG < 10 \text{ then } FG' = (5 \times FG - \min(A,B,C,D))/4$$

Practice session #1: *my personal business card*

We start by converting a downloaded archive into a Git repository to be shared on GitHub.

Step 1: My local Git repository

1. Download the [business card template](#),
2. Unpack the downloaded archive as a local folder,
3. Initialize git inside the folder (by executing **git init** in the local folder, cf. above resources),
4. Compile the card using **xelatex** (cf. [Readme.md](#), install [TeX Live](#) if needed),
 1. If you use the University/School computers, use **pdflatex** instead of xelatex, you may need to comment out lines 7 & 20 to compile.
 2. Disable the fancy font if the compiler fails (i.e., comment out line 8),
 3. Update the file card.tex to add your own details,
5. Recompile the business card to check the rendering.
6. Run **git status**, to list the candidate files for LaTeX,
7. Run **git add YOUR_FILES** to add the source files (card.tex and logo.pdf) to your repository,
8. Commit your changes,
9. Run **git log**,
 - Do you recognise the concepts introduced in the tutorial (commit hash, branches, HEAD)
10. Create a new file named .gitignore and copy the template generated from [gitignore.io](#), ignoring LaTeX files,
11. Commit the new file into your local repository,
12. Run **git status** and compare the output to the previous one,
13. Make sure that no compiled file is tracked by git (e.g., card.pdf),

14. Create a branch new_logo and switch to it,
15. Update the file logo.pdf (e.g., with your custom [avatar](#)),
16. Recompile the card to check how your new logo displays,
17. Commit the changes in the branch new_logo,
18. Switch to the branch master,
19. Compile the card and check its rendering,
20. Merge the branch new_logo into master (you may have to [fix content conflicts](#) or [binary conflicts](#)),
21. Compile the card and check its rendering,
 - Did you get all your changes applied?

Step 2: My remote GitHub repository

1. [Create your own account on GitHub](#) (if you don't have one already),
 - (by using your student email, you can get access to the [GitHub student pack](#))
2. Read the GitHub tutorial here: https://kbroman.org/github_tutorial/pages/init.html
3. Upload your business card to GitHub using the above tutorial,
 - Only source files should be tracked online.
4. Share your GitHub repository with user rouvoy (Settings > Collaborators > Add people).
5. Add your name and GitHub ID to the table below

Step3: *online Git tutorial*

- Online tutorial about Git: <https://learngitbranching.js.org>
- Atlassian's resources for Git: <https://www.atlassian.com/git/tutorials>